Roots of Polynomials

Questions 1–72

Question 1. Each of the following quadratic equations have roots α and β. In each case, find the value of $\alpha + \beta$ and $\alpha\beta$

a) $x^2 + 6x - 7 = 0$
b) $x^2 - 8x + 12 = 0$
c) $2z^2 + 3z + 9 = 0$
d) $-3x^2 + 7x + 8 = 0$
e) $x + 9 = 2x(x - 5)$

Question 2. It is given that

$$ax^2 + bx + c = 0$$

where a, b and c are integers and $a > 0$

a) Given that $c = 0$, find the values of α and β in terms of a and b
b) Given instead that $b = 0$, find the values of α and β in terms of a and c
c) Given instead that $a = 1$, find the values of α and β in terms of b and c

Question 3. Show that:

a) $\frac{1}{\alpha}+\frac{1}{\beta}=\frac{\alpha+\beta}{\alpha\beta}$

b) $\alpha^2+\beta^2=(\alpha+\beta)^2-2\alpha\beta$

c) $\frac{\alpha}{\beta}+\frac{\beta}{\alpha}=\frac{(\alpha+\beta)^2-2\alpha\beta}{\alpha\beta}$

d) $\frac{\alpha^2}{\beta}+\frac{\beta^2}{\alpha}=\frac{(\alpha+\beta)^3-3\alpha\beta(\alpha+\beta)}{\alpha\beta}$

Question 4. Given that

$$2x^2+5x-9=0$$

has roots α and β, without solving the equation find the value of

a) $\alpha+\beta$

b) $\alpha\beta$

c) $\frac{1}{\alpha}+\frac{1}{\beta}$

d) $\alpha^2+\beta^2$

e) $\frac{\alpha}{\beta}+\frac{\beta}{\alpha}$

f) $\frac{\alpha^2}{\beta}+\frac{\beta^2}{\alpha}$

Question 5. Given that

$$7z^2 - 2z + 10 = 0$$

has roots α and β, without solving the equation find the value of:

a) $\alpha + \beta$
b) $\alpha\beta$
c) $\frac{1}{\alpha} + \frac{1}{\beta}$
d) $\alpha^2 + \beta^2$
e) $\frac{\alpha}{\beta} + \frac{\beta}{\alpha}$
f) $\frac{\alpha^2}{\beta} + \frac{\beta^2}{\alpha}$

Question 6. A quadratic equation

$$z^2 + 3z + 10 = 0$$

has roots α and β. Without solving the equation, find a quadratic equation with integer coefficients that has roots

a) α^2 and β^2
b) $\frac{1}{\alpha}$ and $\frac{1}{\beta}$

Question 7. A quadratic equation

$$x^2 - px + q = 0$$

where p and q are positive integers has roots α and β. Given that

$$\alpha + \beta = 10 \text{ and } \alpha^2 + \beta^2 = 94$$

find the values of p and q

Question 8. Given that

$$2x^2 + kx - 3 = 0$$

has roots α and β and k is an unknown integer, without solving the equation find in terms of k the value of each of the following:

a) $\alpha + \beta$
b) $\alpha\beta$
c) $\frac{1}{\alpha} + \frac{1}{\beta}$
d) $\alpha^2 + \beta^2$

Question 9. The equation

$$3x^2 - 2x + k = 0$$

where k is a positive integer has roots α and 3α. Find the value of k

Question 10. The equation

$$4x^2 - (9 + k)x + k = 0$$

where k is a positive integer has roots α and $\alpha + 3$. Find the possible values of k

Question 11. The two equations

$$x^2 - 7x - p = 0 \text{ and } x^2 + 9x + p = 0$$

where p is a positive constant have a common root. Find the value of p

Question 12. The quadratic equation

$$2x^2 + kx + 5 = 0$$

where k is an integer has roots α and β
Find $(\alpha - \beta)^2$ in terms of k

Question 13. The quadratic equation

$$x^2 - 7x + 10 = 0$$

has roots α and β. Without finding the value of α or β show that $\alpha^3 + \beta^3 = 133$

Question 14. The quadratic equation

$$z^2 - 4z + 5 = 0$$

has roots α and β. Without finding the value of α or β show that $\alpha^4 + \beta^4 = -14$

Prerequisite knowledge for Q15: Geometric Sequences

Question 15. The roots of the equation

$$ax^2 + bx + c = 0$$

where a, b and c are positive integers form the first two terms of a geometric series with common ratio of 3. Prove that

$$3b^2 = 16ac$$

Prerequisite knowledge for Q16: Arithmetic Sequences

Question 16. The roots of the equation

$$ax^2 + bx + c = 0$$

where a, b and c are positive integers form the first two terms of a geometric series with common difference of 4. Prove that

$$b^2 - 16a^2 = 4ac$$

Question 17. The quadratic equation

$$x^2 - 6x + 8 = 0$$

has roots α and β.
Without finding the value of α or β, find a quadratic equation with integer coefficients which has roots $\alpha^3 + \beta$ and $\beta^3 + \alpha$

Question 18. Two functions f and g are defined to be

$$f(x) = px^2 - 24x + 17$$

and

$$g(x) = qx^2 - 4x + 3$$

where p and q are real constants.
It is known that the sum of the roots of f is equal to the product of the roots of g. Find the relationship between p and q

Question 19. The roots of

$$x^2 + px + q = 0$$

are α and β where $\alpha = 2\beta$
Show that $9q = 2p^2$

Prerequisite knowledge for Q20: Complex Numbers

Question 20. The roots of

$$z^2 - kz + 25 + 18i = 0$$

where k is a complex number, are α and $5 + i$. Find the value of k

Question 21. The roots of

$$ax^2 + bx + c = 0$$

where a, b, c are positive integers, are α and β

a) Find a quadratic equation with integer coefficients which has roots

$$\alpha + \beta \text{ and } \alpha\beta$$

in terms of a, b and c

b) Show that the equation you found in part a) will always have two real distinct roots.

Question 22. For the integer k, the quadratic equation

$$x^2 - 6x + k = 0$$

has roots α and β where $\alpha > \beta$

a) Write down $\alpha\beta$ in terms of k

b) Write down the value of $\alpha + \beta$

c) Show that a quadratic equation with roots

$$\alpha - 2 \text{ and } \beta + 2$$

is given by

$$x^2 - 6x + \left(k - 4 + 2\sqrt{36 - 4k}\right) = 0$$

Cubics

Question 23. Each of the following cubic equations have roots α, β and γ. In each case, find the value of $\alpha+\beta+\gamma$ and $\alpha\beta+\alpha\gamma+\beta\gamma$ and $\alpha\beta\gamma$

a) $x^3+3x^2-5x-1=0$
b) $x^3-9x^2+5x+4=0$
c) $2z^3+5z^2+11z-2=0$
d) $-5z^3+z^2-12=0$
e) $x+1=4x(x-1)(x+2)$

Question 24. A cubic equation $f(z)=0$ has roots α, β and γ
It is known that

$$\alpha+\beta+\gamma=3,\ \alpha\beta+\alpha\gamma+\beta\gamma=8$$

and $\alpha\beta\gamma=-4$
Write down an expression for $f(z)$ given that the coefficient of the z^3 term is 2

Question 25. A cubic equation

$$x^3 + 6x^2 - px - q = 0$$

where p and q are positive integers has roots 2, -1 and α

a) Find the value of α

b) Find the values of p and q

Prerequisite knowledge for Q26: Complex Numbers

Question 26. A cubic equation is given by

$$2z^3 + pz^2 - 4z + q = 0$$

where p and q are positive integers, the sum of the roots is $-\frac{3}{2}$ and the product of the roots is -10

a) Find the values of p and q

b) Use your calculator to find all the roots of the equation giving your answer to 3 decimal places.

Question 27. Show that

a) $\frac{1}{\alpha}+\frac{1}{\beta}+\frac{1}{\gamma}=\frac{\alpha\beta+\alpha\gamma+\beta\gamma}{\alpha\beta\gamma}$

b) $\alpha^2+\beta^2+\gamma^2=(\alpha+\beta+\gamma)^2-2(\alpha\beta+\alpha\gamma+\beta\gamma)$

c) $(\alpha+\beta)(\alpha+\gamma)(\beta+\gamma)=(\alpha+\beta+\gamma)(\alpha\beta+\alpha\gamma+\beta\gamma)-\alpha\beta\gamma$

d) $\alpha^2\beta^2+\alpha^2\gamma^2+\beta^2\gamma^2=(\alpha\beta+\alpha\gamma+\beta\gamma)^2-2\alpha\beta\gamma(\alpha+\beta+\gamma)$

Question 28. Find each of the following in terms of $\alpha+\beta+\gamma$ and $\alpha\beta+\alpha\gamma+\beta\gamma$ and $\alpha\beta\gamma$

a) $\alpha^2\beta\gamma+\alpha\beta^2\gamma+\alpha\beta\gamma^2$

b) $\alpha^2\beta^2+\alpha^2\gamma^2+\beta^2\gamma^2$

Question 29. Given that

$$3x^3 + 4x^2 - 8x - 2 = 0$$

has roots α, β and γ, without solving the equation find the value of

a) $\alpha + \beta + \gamma$
b) $\alpha\beta + \alpha\gamma + \beta\gamma$
c) $\alpha\beta\gamma$
d) $\frac{1}{\alpha} + \frac{1}{\beta} + \frac{1}{\gamma}$
e) $\alpha^2 + \beta^2 + \gamma^2$
f) $\alpha^2\beta\gamma + \alpha\beta^2\gamma + \alpha\beta\gamma^2$
g) $\alpha^2\beta^2 + \alpha^2\gamma^2 + \beta^2\gamma^2$
h) $(\alpha + \beta)(\alpha + \gamma)(\beta + \gamma)$
i) $\frac{\alpha^2\beta^2 + \alpha^2\gamma^2 + \beta^2\gamma^2}{\alpha\beta\gamma}$

Question 30. Given that

$$5z^3 - 2z^2 + z + 1 = 0$$

has roots α, β and γ, without solving the equation find the value of

a) $\alpha + \beta + \gamma$
b) $\alpha\beta + \alpha\gamma + \beta\gamma$
c) $\alpha\beta\gamma$
d) $\frac{1}{\alpha} + \frac{1}{\beta} + \frac{1}{\gamma}$
e) $\alpha^2 + \beta^2 + \gamma^2$
f) $\alpha^2\beta\gamma + \alpha\beta^2\gamma + \alpha\beta\gamma^2$
g) $\alpha^2\beta^2 + \alpha^2\gamma^2 + \beta^2\gamma^2$
h) $(\alpha + \beta)(\alpha + \gamma)(\beta + \gamma)$
i) $\frac{\alpha^2\beta^2 + \alpha^2\gamma^2 + \beta^2\gamma^2}{\alpha\beta\gamma}$

Question 31. The cubic equation

$$x^3 + px + q = 0$$

where p and q are real has roots α, β and γ
Explain why $\alpha + \beta = -\gamma$

Question 32. The cubic equation

$$x^3 + px^2 + 5p = 0$$

where p is an integer has roots α, β and γ
The product of two of the roots is 12
Show that the final root is $-\frac{5p}{12}$

Question 33. The equation

$$6x^3 + kx^2 - 26x + 15 = 0$$

where k is a real number has roots α, β and γ
Find the value of k when the roots sum to $\frac{11}{6}$

Question 34. For positive integer k, the equation

$$10x^3 - 57x^2 + kx - 5 = 0$$

has roots α, $\frac{1}{\alpha}$ and β

a) Solve the equation.

b) Find the value of k

Question 35. Given that

$$z^3 - 4z^2 + 3z - 7 = 0$$

has roots α, β and γ, without solving the equation find the value of

a) $\frac{2}{\alpha} + \frac{2}{\beta} + \frac{2}{\gamma}$

b) $(\alpha + 3)(\beta + 3)(\gamma + 3)$

c) $\frac{\alpha^2\beta^2 + \alpha^2\gamma^2 + \beta^2\gamma^2}{5\alpha\beta\gamma}$

d) $(2\alpha + 2\beta)(2\alpha + 2\gamma)(2\beta + 2\gamma)$

Question 36. For the cubic equation

$$x^3 + kx^2 - 5x + 3 = 0$$

with roots α, β and γ, show that

$$\alpha^2\beta^2 + \alpha^2\gamma^2 + \beta^2\gamma^2 = 25 - 6k$$

Question 37. The function f is defined by

$$f(z) = z^3 - 7z^2 + kz - 120$$

where k is a real constant.
Given that the roots of the equation $f(z) = 0$ are α, β and γ and that

$$\alpha^2 + \beta^2 + \gamma^2 = -55$$

a) Explain why $f(z) = 0$ has only one real root.
b) Find the value of k

Prerequisite knowledge for Q38-Q39: Complex Numbers

Question 38. Given

$$z^3 + kz + 35 - 5i = 0$$

where k is a complex number has roots α, β and γ. It is known that

$$\alpha + \beta = 5$$

Find the value of k

Question 39. Given

$$z^3 + kz^2 + 6 + i = 0$$

where k is a complex number has roots α, β and γ.
It is known that

$$\alpha^2 + \beta^2 + \gamma^2 = 2i$$

Find the possible values of k

Question 40. Given that

$$2x^3 + 5x^2 - 4 = 0$$

has roots α, β and γ.

a) By considering

$$(\alpha + \beta + \gamma)^3$$

and using

$$\begin{aligned}\alpha^2\beta + \alpha^2\gamma + \alpha\gamma^2 + \alpha\beta^2 + \beta^2\gamma + \beta\gamma^2 + 2\alpha\beta\gamma \\ = (\alpha + \beta + \gamma)(\alpha\beta + \alpha\gamma + \beta\gamma) - \alpha\beta\gamma\end{aligned}$$

find

$$\alpha^3 + \beta^3 + \gamma^3$$

in terms of $\alpha + \beta + \gamma$, $\alpha\beta + \alpha\gamma + \beta\gamma$ and $\alpha\beta\gamma$

b) Hence, without solving the equation find the value of

$$\alpha^3 + \beta^3 + \gamma^3$$

Question 41. The cubic equation

$$2x^3 + 7x^2 - 28x + 12 = 0$$

has roots α, β and γ
The cubic equation

$$z^3 + pz^2 + qz + r = 0$$

where p, q and r are integers, has roots $\alpha\beta$, $\beta\gamma$ and $\alpha\gamma$
Find the values of p, q and r

Question 42. It is known that the cubic equation

$$x^3 + px^2 + qx - 100 = 0$$

where p and q are real constants, has solutions

$$\alpha, \frac{20}{\alpha} \text{ and } \alpha - 5$$

a) Find the values of the solutions to the equation.
b) Find the values of p and q

Question 43. It is known that the cubic equation

$$x^3 + px^2 + qx - 216 = 0$$

where p and q are real constants has solutions

$$\alpha, \frac{36}{\alpha} \text{ and } \alpha + 4$$

a) Find the values of the solutions to the equation.
b) Find the values of p and q

Question 44. A cubic equation

$$36x^3 - 444x^2 + px - q = 0$$

where p and q are positive integers has roots

$$\alpha, \frac{1}{\alpha} \text{ and } \alpha + \frac{1}{\alpha}$$

a) Find the value of α and state the roots.
b) Find the value of p and q

Question 45. Given the cubic equation

$$2x^3 + 3x^2 - 7x + 2 = 0$$

has roots α, β and γ,

a) determine the value of

$$\frac{1}{\alpha} + \frac{1}{\beta} + \frac{1}{\gamma}$$

b) Find a cubic equation with integer coefficients which has roots $\frac{1}{\alpha}$, $\frac{1}{\beta}$ and $\frac{1}{\gamma}$

Question 46. A cubic equation

$$3z^3 - 5z^2 + 7z - 4 = 0$$

has roots α, β and γ.
Without solving the equation, find a cubic equation with integer coefficients that has roots

a) $\alpha\beta$, $\alpha\gamma$ and $\beta\gamma$

b) $\alpha + \beta$, $\beta + \gamma$ and $\gamma + \alpha$

Question 47. Given that

$$3x^3 + 4x^2 - 8x + k = 0$$

where k is a positive constant has roots α, β and γ, find the value of

$$(\alpha + \beta)(\alpha + \gamma)(\beta + \gamma)$$

giving your answer in terms of k

Question 48. The cubic equation

$$px^3 + 11x^2 - 106x + q = 0$$

where p and q are positive integers has roots

$$\alpha, \beta \text{ and } \gamma$$

The cubic equation

$$20w^3 + 93w^2 - rw - 685 = 0$$

where r is a positive constant has roots

$$(3\alpha - 1), (3\beta - 1) \text{ and } (3\gamma - 1)$$

Without solving either cubic equation, find the values of p, q and r

Question 49. The cubic equation

$$x^3 + 10x^2 + kx - 126 = 0$$

where k is an integer has roots α, β and γ
Given that

$$\alpha^2 + \beta^2 + \gamma^2 = 94$$

a) find the value of k
b) Find a cubic equation with integer coefficients which has roots 2α, 2β and 2γ

Question 50. The cubic equation

$$z^3 - pz^2 + 4z + q = 0$$

where p and q are positive integers has roots α, β and γ
Given that

$$\alpha^2 + \beta^2 + \gamma^2 = 17$$

a) Determine the value of p

Given that also

$$\frac{1}{\alpha} + \frac{1}{\beta} + \frac{1}{\gamma} = -\frac{2}{3}$$

b) Determine the value of q
c) Without solving the equation, determine the value of

$$(\alpha + 1)(\beta + 1)(\gamma + 1)$$

Prerequisite knowledge for Q51: Complex Numbers

Question 51. Two functions f and g are defined as

$$f(z) = z^3 - pz^2 - 4$$

and

$$g(z) = 3z^3 - 12z^2 + qz - 12$$

where p and q are real constants.
It is known that the sum of the roots of f is equal to the product of the roots of g

a) State the value of p

Given that a root of g is $1 - i$

b) Find the value of q

Question 52. The equation

$$3x^3 + kx^2 + 85x - 75 = 0$$

where k is an integer has roots α, β and $\frac{\alpha}{\beta}$

a) Find the possible solutions of the equation

b) Find the possible values of k

Question 53. The equation

$$x^3 + kx^2 + 7 = 0$$

has roots α, β and γ.
The equation

$$8x^3 - 32x^2 + px + q = 0$$

has roots $\alpha\beta + \gamma$, $\beta\gamma + \alpha$ and $\alpha\gamma + \beta$
Show that $k = -4$

Question 54. The function f is defined by

$$f(z) = z^3 - az^2 - 13z + b = 0$$

where a and b are real positive constants has roots α, β and -4
Show that

$$b = 12 + 16a$$

Prerequisite knowledge for Q55: Complex Numbers

Question 55. The equation

$$z^3 - 4z^2 + kz - 8 = 0$$

where k is a positive integer has roots α, β and $\alpha - \beta + \frac{8}{\alpha}$

a) Find the solutions to the equation.

b) Find the possible values of k

Question 56. For the integer p, the cubic equation

$$x^3 - (p+4)x^2 + 3px - 9p = 0$$

has roots α, β and γ. Show that the sum of the square of the roots is

$$p^2 + 2p + 16$$

Question 57. For the integer p, the cubic equation

$$x^3 - (p+6)x^2 + (5p+4)x - (7p+3) = 0$$

has roots α, β and γ. Show that the sum of the products of the squares of pairs of roots is

$$11p^2 - 50p - 20$$

Question 58. The cubic equation

$$x^3 - (p+3)x^2 + 4px - 12p = 0$$

has roots α, β and γ. Show that the sum of the cube of the roots is

$$p^3 - 3p^2 + 27p + 27$$

Question 59. The roots of

$$x^3 + px^2 + qx + r = 0$$

are α, β and γ.
Given that $\alpha = \beta = \gamma$ show that $p = \frac{9r}{q}$

Prerequisite knowledge for Q60-Q61: Geometric Sequences

Question 60. The roots of the equation

$$x^3 - 49x^2 - 1029x + k = 0$$

where k is a positive integer form a geometric sequence. Find the value of k

Question 61. The roots of the equation

$$ax^3 + bx^2 + cx + d = 0$$

form a geometric sequence. Prove that

$$ac^3 - db^3 = 0$$

Prerequisite knowledge for Q62-Q63: Arithmetic Sequences

Question 62. The roots of the equation

$$2x^3 + 6x^2 - 26x - k = 0$$

where k is a positive integer form an arithmetic sequence. Find the value of k

Question 63. The roots of the equation

$$px^3 + qx^2 + rx + s = 0$$

form an arithmetic sequence. By using $\alpha - d$, α and $\alpha + d$ as the roots prove that

$$2q^3 + 27p^2s = 9pqr$$

Question 64. Given that α, β and γ are real roots of the cubic equation

$$x^3 + px^2 + qx + r = 0$$

prove that $q^2 \geq 2pr$

Quartics

Question 65. Each of the following quartic equations have roots α, β, γ and δ. In each case, find the value of $\sum \alpha$, $\sum \alpha\beta$, $\sum \alpha\beta\gamma$ and $\sum \alpha\beta\gamma\delta$

a) $z^4 + 5z^3 - 3z^2 + 6z - 2 = 0$
b) $z^4 - 4z^3 + 4z^2 - z + 8 = 0$
c) $2x^4 + 5x^3 + 6x^2 + 11x + 1 = 0$
d) $-4z^4 - 4z^3 + 7z^2 + 20z - 2 = 0$
e) $z^4 + 7 = z^2(z - 4)$

Question 66. Given that

$$8z^4 + z^3 - 3z^2 + 5z - 7 = 0$$

has roots α, β, γ and δ, without solving the equation find the value of

a) $\alpha + \beta + \gamma + \delta$
b) $\alpha\beta + \alpha\gamma + \alpha\delta + \beta\gamma + \beta\delta + \gamma\delta$
c) $\alpha\beta\gamma + \alpha\gamma\delta + \alpha\beta\delta + \beta\gamma\delta$
d) $\alpha\beta\gamma\delta$
e) $\frac{1}{\alpha} + \frac{1}{\beta} + \frac{1}{\gamma} + \frac{1}{\delta}$
f) $\alpha^2 + \beta^2 + \gamma^2 + \delta^2$
g) $\alpha^2\beta^2\gamma\delta + \alpha^2\beta\gamma^2\delta + \alpha^2\beta\gamma\delta^2 + \alpha\beta^2\gamma^2\delta + \alpha\beta^2\gamma\delta^2 + \alpha\beta\gamma^2\delta^2$
h) $\frac{2}{\alpha\beta\gamma} + \frac{2}{\alpha\beta\delta} + \frac{2}{\alpha\gamma\delta} + \frac{2}{\beta\gamma\delta}$

Question 67. Given that

$$12z^4 + 3z^3 - 2z^2 - 4z + 3 = 0$$

has roots α, β, γ and δ, without solving the equation find the value of

a) $\alpha + \beta + \gamma + \delta$
b) $\alpha\beta + \alpha\gamma + \alpha\delta + \beta\gamma + \beta\delta + \gamma\delta$
c) $\alpha\beta\gamma + \alpha\gamma\delta + \alpha\beta\delta + \beta\gamma\delta$
d) $\alpha\beta\gamma\delta$
e) $\frac{1}{\alpha} + \frac{1}{\beta} + \frac{1}{\gamma} + \frac{1}{\delta}$
f) $\alpha^2 + \beta^2 + \gamma^2 + \delta^2$
g) $\alpha^2\beta^2\gamma\delta + \alpha^2\beta\gamma^2\delta + \alpha^2\beta\gamma\delta^2 + \alpha\beta^2\gamma^2\delta + \alpha\beta^2\gamma\delta^2 + \alpha\beta\gamma^2\delta^2$
h) $\frac{2}{\alpha\beta\gamma} + \frac{2}{\alpha\beta\delta} + \frac{2}{\alpha\gamma\delta} + \frac{2}{\beta\gamma\delta}$

Question 68. Given that

$$2z^4 + 3z^3 - 2z + 3 = 0$$

has roots α, β, γ and δ, without solving the equation find the value of

a) $\frac{3}{\alpha} + \frac{3}{\beta} + \frac{3}{\gamma} + \frac{3}{\delta}$
b) $\alpha^2 + \beta^2 + \gamma^2 + \delta^2$
c) $(1 - \alpha)(1 - \beta)(1 - \gamma)(1 - \delta)$

Question 69. A quartic equation

$$6x^4 - 35x^3 + 62x^2 - px + q = 0$$

where p and q are positive integers has roots α, $\frac{1}{\alpha}$, β and $\frac{1}{\beta}$

a) Find the value of q

b) Find the value of p

Question 70. A quartic equation

$$2x^4 + 3x^3 + px^2 + qx + 882 = 0$$

where p and q are integers has real roots α, 2α, β and $\frac{\beta}{2}$

a) Find the value of the roots.

b) Find the values of p and q

Question 71. The equation

$$9x^4 - 150x^3 + px^2 - qx + 63 = 0$$

where p and q are positive integers has roots α, β, γ and δ
Given that two of the roots have the same value and the product of the other two is 63, and all the roots have positive values,

a) find the solutions to the equation.
b) Find the values of p and q

Prerequisite knowledge for Q72: Geometric Sequences

Question 72. The quartic equation

$$x^4 + bx^3 + cx^2 + dx + e = 0$$

has roots α, β, γ and δ
The roots form a geometric series with common ratio k
Show that

$$c = \alpha^2 \left[k^3 + \sum_{r=1}^{n} k^r \right]$$

where n is an integer to be found.